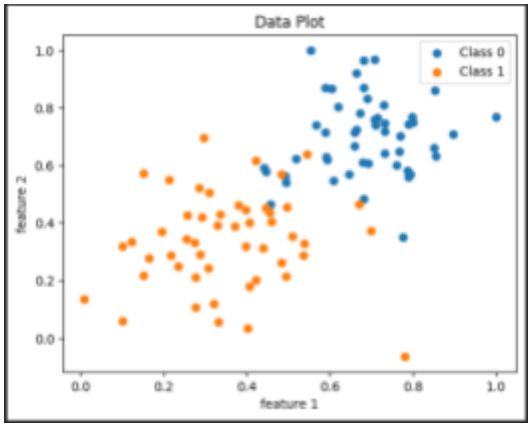
Assign 6 Report

Initial Setup:

Data file is called "data.csv" in same directory as notebook.

figure below: visualize data as scatterplot



Part one: Step function

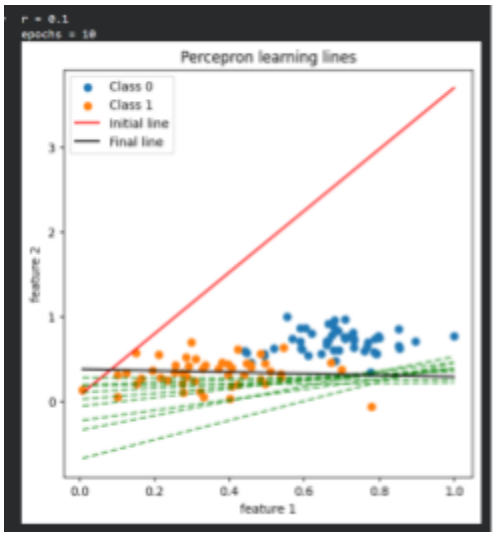
figure below: code snippet of prediction logic including step function classification

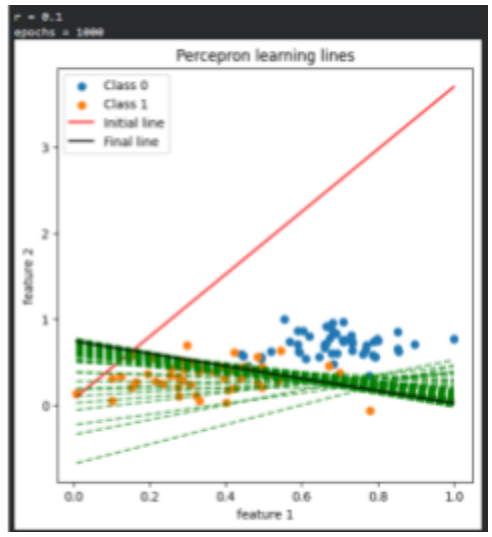
```
def preceptron_step(x, w, b):
  wx_product = 0

  #calculate weight and feature products in linear combination
  for i in range(len(w)):
    wx_product += w[i] * x[i]

  #add bias
  predict = wx_product + b

  #classify prediction using a step function
  if predict >= 0: #check sign of prediction
    return 1
  else:
    return 0
```

Experimenting with learning rate and epochs:

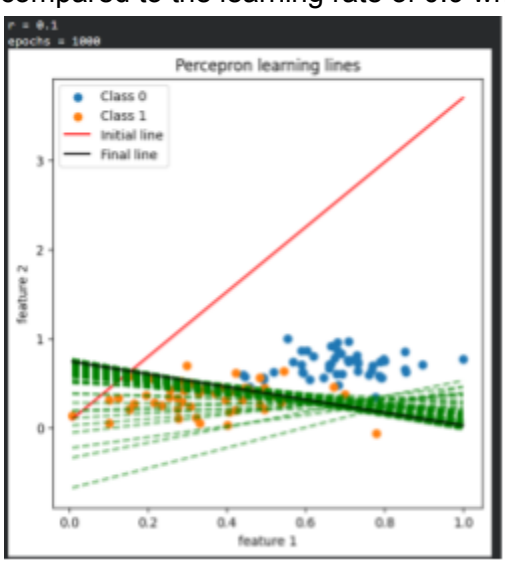
With learning rate 0.1 and increased epochs from 10 to 1000
Ten iterations at this learning rate did not settle to a stable value. Many points that should be in class 1 are misclassified. With a huge increase in iterations, 1000 shows a better line history. The weights and bias are updated less often as it converges. Some class 1 points are misclassified, but it is error is reduced compared to only 10 iterations.

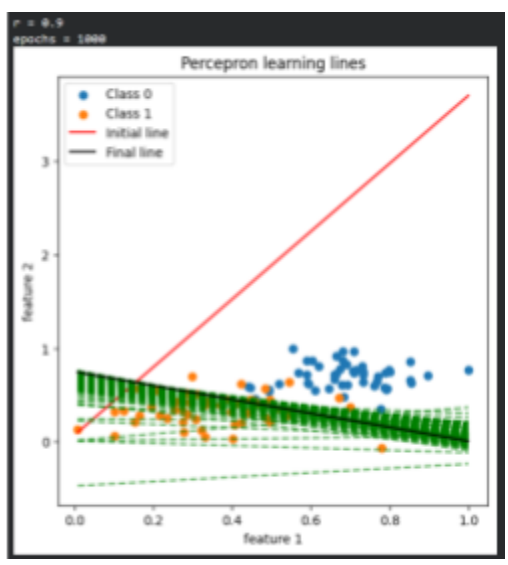
Assign 6 Report





With epochs 1000 and increased learning rate from 0.1 to 0.9
When the learning rate is 0.1 the weights and bias are updated in smaller increments compared to the learning rate of 0.9 where the error oscillates more before converging.





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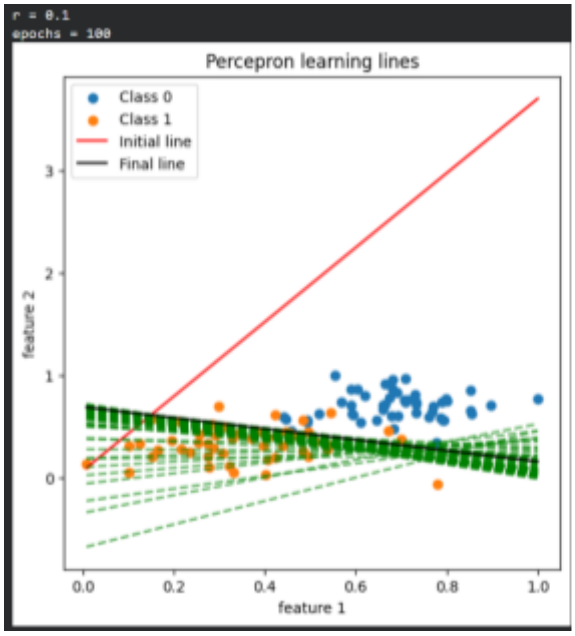
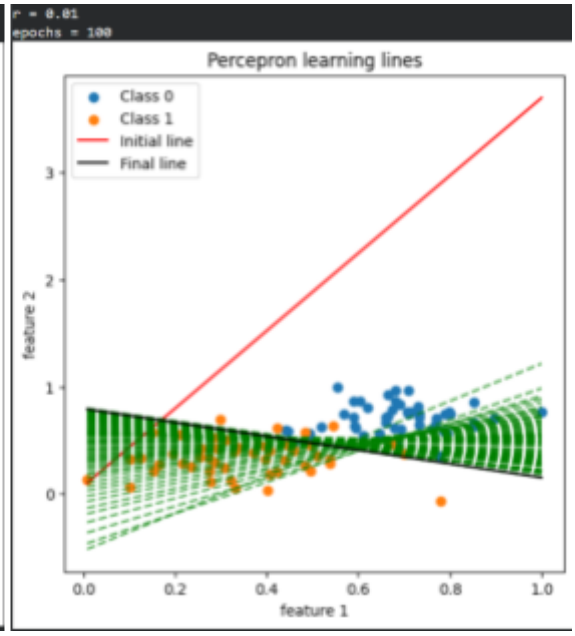
With epochs 100 and decreased learning rate from 0.1 to 0.01
With even smaller updates the weights and bias the error starts to converge after 100 epochs. The lower learning rate does not oscillate before it settles to a minima.





Part 2: Sigmoid function

code snippet of gradient decent version

```
def preceptron_sig(x, w, b):
  wx_product = 0

  #calculate weight and feature products in linear combination
  for i in range(len(w)):
    wx_product += w[i] * x[i]

  #add bias
  predict_grdt = wx_product + b

  #classify prediction using a sigmoid function
  return 1 / (1 + np.exp(-predict_grdt))
```

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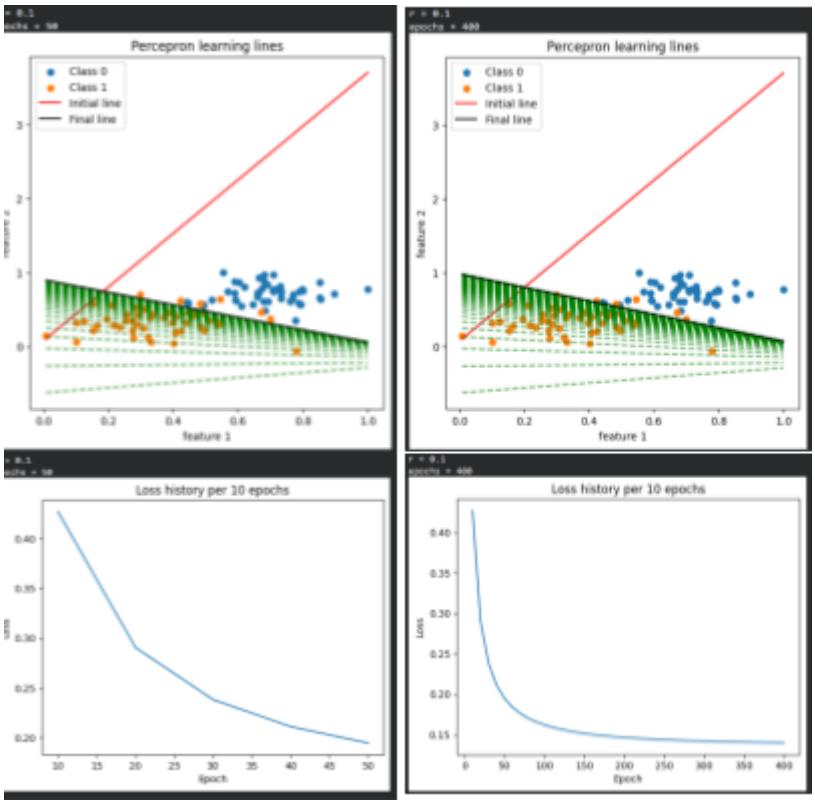
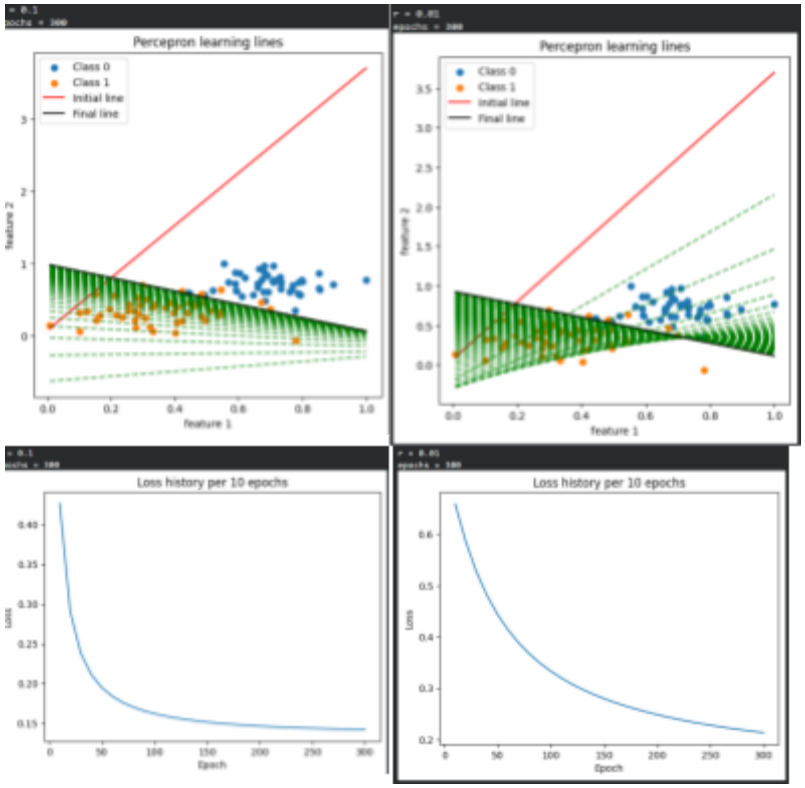
Experimenting with learning rate and epochs:

With learning rate 0.1 and increased epochs from 50 to 400
By increasing the epochs the error is reduced further. More points are accurately classified.

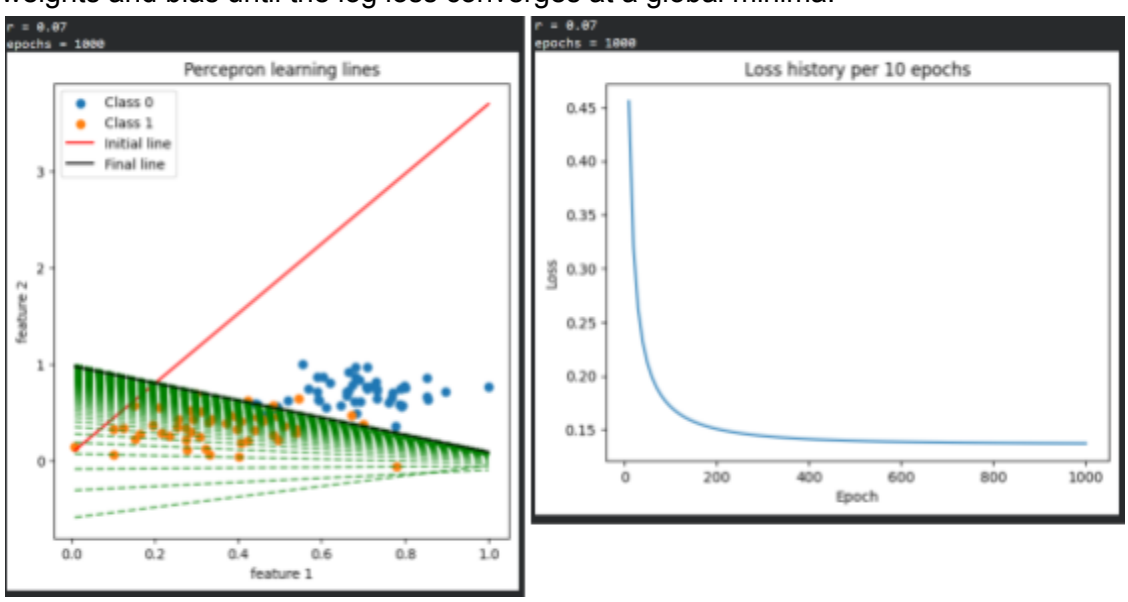


With epochs 300 and decreased learning rate from 0.1 to 0.01
The decreasing the learning rate greatly did not reduce the error. Only 300 iterations was not enough to adjust the weights far enough from their random initial value. A learning rate lower than 0.1 will need more than 300 epochs to converge to a global minima.

Assign 6 Report



With learning rate from 0.07 increasing epochs from 400 to 1000
These parameters achieve the best error reduction out of above figures. With a learning rate low enough to converge without oscillating and enough iterations to update the weights and bias until the log loss converges at a global minima.



Assign 6 Report

Disclosure of AI tools and reused code:

- Google Colab on board Gemini was used to debug syntax errors and file input errors caused by missing headers in the data.csv file. ChatGPT was used to assist with translating some c++ logic into Python, as well as the syntax and uses of functions such as numpy.clip() and plotting in Matplotlib.
- Code from Assignment 4 starter code was reused for visualization logic.